

Xuangeng CHU

xuangeng.chu@mi.t.u-tokyo.ac.jp | xg.chu@outlook.com

Research Interests

1. Novel view synthesis (NeRF / 3DGS).
2. Human-centric 3D vision, digital human, shape / pose generation and reconstruction.

Education

The University of Tokyo

Ph.D. Candidate in RCAST, University Fellowship, Supervisor: Prof. Tatsuya Harada Tokyo, Japan

Peking University

M.Eng. in Software Engineering, Supervisor: Prof. Yasha Wang Beijing, China

Tongji University

B.Eng. in Computer Science Shanghai, China

Work Experience

Tencent, Research Engineer

Applied Research Center Mar 2021 – Oct 2022 Shenzhen, China

- Worked on category extensible object detection algorithm for videos.
- Worked on articulated model reconstruction algorithm for a virtual scene generating system.

Internship Experience

Princeton University, Visiting student researcher

Advised by Prof. Jia DENG. Jul 2024 – Aug 2024 New Jersey, USA

Worked on research of general one-shot 3D reconstruction.

International Digital Economy Academy, Research intern

Advised by Dr. Yu LI Dec 2022 - Nov 2023 Shenzhen, China / Remote

Worked on research and development of human reconstruction and pose estimation.

Microsoft Research Asia, Research intern

Advised by Dr. Xiulian PENG Jun 2020 - Feb 2021 Beijing, China

Worked on research of audio-visual speech separation problem (cocktail party problem).

MEGVII Technology, Research intern

Advised by Dr. Xiangyu ZHANG Jan 2019 - Jun 2020 Beijing, China

Worked on research and development of object detection in crowded scenes.

Publication

Generalizable and Animatable Gaussian Head Avatar
Xuangeng Chu, Tatsuya Harada

NeurIPS 2024

We propose the first generalizable 3DGS head avatar framework that achieves single forward reconstruction and real-time reenactment. The key idea of this work is lifting two sets of 3D Gaussian points from the input image. Project website and code: https://xg-chu.github.io/project_gaga.

GPAvatar: Generalizable and Precise Head Avatar from Image(s) **ICLR 2024**

Xuangeng Chu, Yu Li, Ailing Zeng, Tianyu Yang, Lijian Lin, Yunfei Liu, Tatsuya Harada

We propose a framework to reconstructs 3D head avatars from one or several images in a single forward pass. The key idea of this work is a dynamic point-based expression field and a attention-based fusion module. Project website and code: https://xg-chu.github.io/project_gpavatar.

Accurate 3D Face Reconstruction with Facial Component Tokens **ICCV 2023**

Tianke Zhang, **Xuangeng Chu**, Yunfei Liu, Lijian Lin, Zhendong Yang, Zhengzhuo Xu, Chengkun Cao, Fei Yu, Changyin Zhou, Chun Yuan, Yu Li

We propose a framework for 3D face reconstruction from monocular images based on 3DMM and transformers. Our method uses separate tokens to improve the disentanglement of shape and expression for more accurate reconstruction.

Detection in Crowded Scenes: One Proposal, Multiple Predictions **CVPR 2020 Oral**

Xuangeng Chu*, Anlin Zheng*, Xiangyu Zhang*, Jian Sun

We propose a simple and almost cost-free method to improve the detection performance in crowded scens. The key idea of this work is to predict a set of instances from each proposal region instead of just one. Code is available on: <https://github.com/xg-chu/CrowdDet>.

Services

Reviewer, TPAMI; CVPR 2022, 2023; ACMMM 2024; NeurIPS 2024.